

HASHWANTH GHANTA

M.Sc. Digital Engineering Candidate | B.Tech Electronics & Communication Engineering

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PORTFOLIO - hashwanthghanta.github.io

3D _ SIMULATION - play.unity.com — [Post-Combustion CO2 Carbon Capture](#)

PROFESSIONAL SUMMARY

Digital Engineering M.Sc. student at OVGU Magdeburg with a strong background in Electronics & Communication Engineering (B.Tech, VIT-AP University). Specialises in data-driven software development, Unity 3D simulation, and C# architecture. Combines technical depth in embedded systems and signal processing with hands-on experience in interactive simulation design and UI/UX development. Seeking a Werkstudent position to apply analytical problem-solving and software engineering skills in real-world industry applications.

EDUCATION

M.Sc. Digital Engineering

April 2025 – Present

Otto von Guericke University Magdeburg (OVGU)

Magdeburg, Germany

- Focusing on data-driven software development, simulation engineering, and embedded systems
- Coursework: Advanced Software Engineering, Digital Systems Design, Human-Computer Interaction

B.Tech Electronics & Communication Engineering

2020 – 2024

VIT-AP University

Amaravati, India

- Cumulative GPA: 7.72 / 10.0
- Relevant coursework: Signal Processing, Embedded Systems, Programming in C/C++, Communication Theory

PROJECTS

CO2 Carbon Capture Process Simulation

2025 – Present

Unity 3D / C# — Interactive Educational Simulation

- Developed a fully interactive Unity 3D educational simulation for a 7-phase post-combustion CO2 carbon capture process, featuring custom C# architecture and real-time dynamic UI panels
- Implemented particle fluid flow animations and responsive 3D environment to visualise industrial CCS technology
- Designed modular scene management and event-driven UI to support step-by-step guided learning experiences
- Live simulation: play.unity.com — [Post-Combustion CO2 Carbon Capture](#)

CO2 Carbon Capture Dashboard App

2025

C# / Data Visualisation

- Built a data-driven dashboard app displaying simulation axis data for the post-combustion CO2 capture process
- Integrated real-time data rendering with a clean UI for analytical review of capture process stages

IoT Smart Helmet & Bike Safety System |

VIT-AP Final Year Project 2024

- Built a GPS + alcohol sensor-equipped smart helmet to detect unsafe riding and alert emergency contacts
- Targeted reduction of motorcycle fatalities through automated safety enforcement

Arduino Blind Spot Detector | VIT-AP Project 2022 India

- Developed an ultrasonic sensor-based blind spot alert system for vehicles to improve driver awareness and reduce collision risk

IoT-Based Ambulance Assistance System |

VIT-AP, 2023 India

- Designed an intelligent IoT traffic junction system giving ambulances top priority clearance.
- Implemented using microcontrollers with real-time sensor detection and relay control

Arduino Blind Spot Detector |

VIT-AP Project 2022 India

- Developed an ultrasonic sensor-based blind spot alert system for vehicles to improve driver awareness and reduce collision risk

TECHNICAL SKILLS

Programming Languages: C, C#, Python, HTML5, JavaScript

Game & Simulation: Unity 3D, C# (MonoBehaviour, ScriptableObjects, Event Systems)

Design & Media: Blender (3D Modelling), Adobe Premiere Pro, Canva

DevOps & Tools: Git, GitHub, GitHub Actions, VS Code, Visual Studio

Domains: Embedded Systems, Signal Processing, Data-Driven Development, UI/UX Design

CERTIFICATIONS & TRAINING

- Unity Certified Associate — Game Developer (in progress)
- Python for Data Science — IBM / Coursera
- Electronics & IoT Fundamentals — NPTEL
- Data Visualisation: Empowering Business with Effective Insights — Tata Group via Forage (2024)
- Career Essentials in Business Analysis — Microsoft & LinkedIn Learning (2024)
- Introduction to Business Analytics: Communicating with Data — Coursera
- IoT Wireless & Cloud Computing Emerging Technologies — Coursera
- Introduction and Programming with IoT Boards — Coursera
- Project Management Foundations — Coursera

LEADERSHIP & ACTIVITIES

Volunteer Head — VITOPIA 2024 | VIT-AP University Annual Cultural Festival 2024 India

- Managed the largest annual college event with 1,000+ participants including students and celebrity guests
- Coordinated logistics, student teams, event scheduling, and live activity management across cultural and technical events